

Modeling Measures for Evaluating LLM Explanation Faithfulness

Introduction

- Explainable artificial intelligence tries to improve AI trust by developing AI systems that either explain themselves or explain other AI systems.
- These explanations provide insight into a model's decision so they can be checked for factuality, consistent reasoning, and adherence to policy.
- An explanation is faithful if it accurately represents the reasoning process behind a model's prediction.
- Faithful explanations help researchers detect hidden bias and spurious correlations.
- There is currently no way to directly measure faithfulness, but 60+ measures proposed.
- The Explanation Faithfulness Evaluation Measures Ontology (EFEMO) organizes existing faithfulness measures.
- It allows comparison of measures to help researchers identify appropriate measures

EFEMO: The Explanation Faithfulness Evaluation Measures Ontology

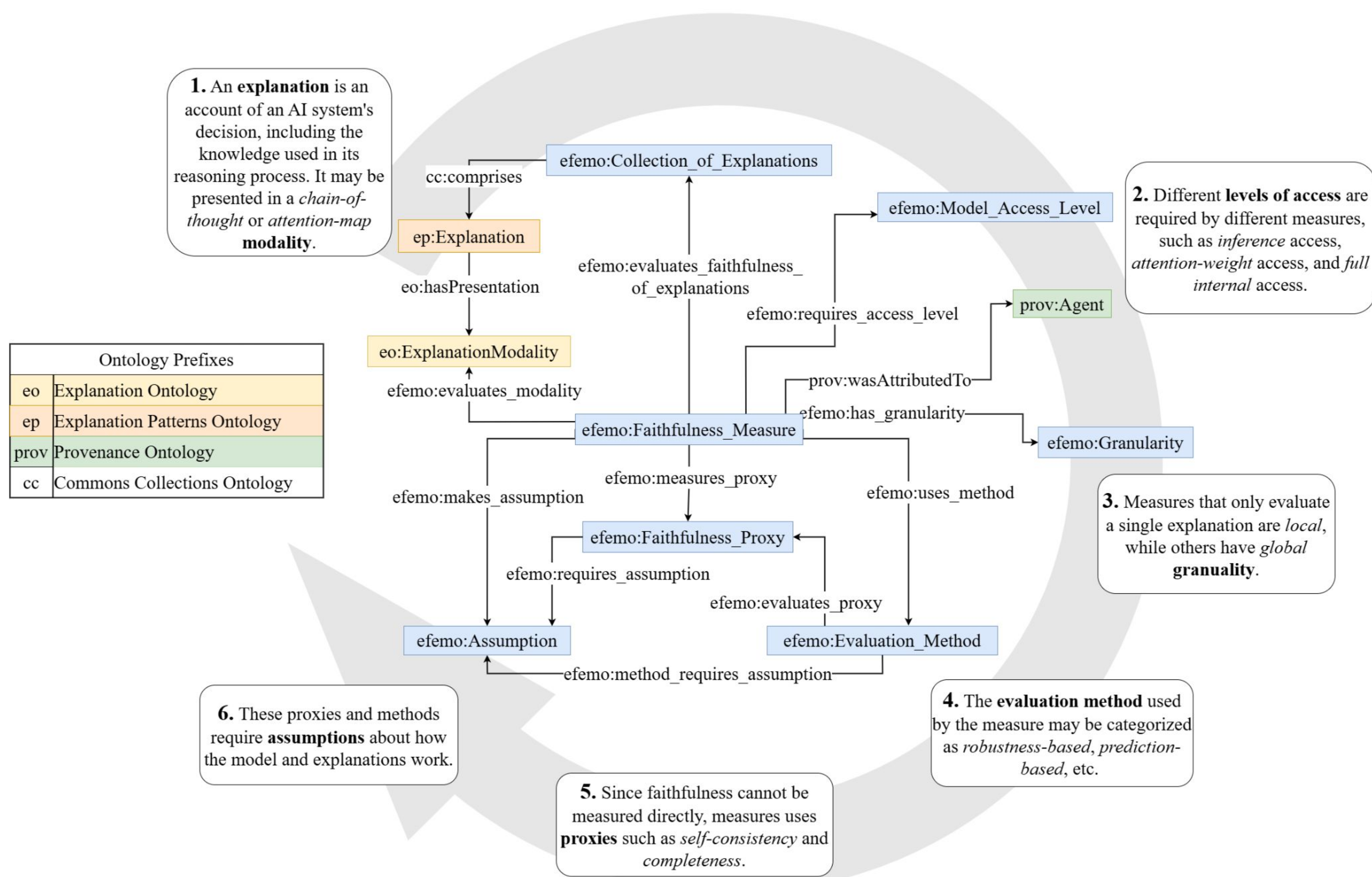


Figure 1: An overview concept diagram of EFEMO.

Ontology Evaluation

Table 1: Competency questions used to validate our ontology

Focus: Evaluation Measures

CQ 1: What article is measure M attributed to?

CQ 2: What explanation modality, granularity, and evaluation method does measure M use?

CQ 3: What faithfulness assumptions are required to measure proxy P?

Focus: Measure Recommendations

CQ 4: What measures evaluate explanation modality D at granularity G using proxy characteristic P?

CQ 5: What measures require model access level L while using an evaluation method of type E?

CQ 6: Are there any measures with the same explanation modality, granularity, and model access requirements as measure M?

Focus: Analysis of Measures

CQ 7: Do measures M and N share any assumptions?

CQ 8: What assumption is most commonly made when evaluating explanations with modality D?

CQ 9: List the proxy characteristics used in evaluations that require model access level L by popularity.

CQ 10: What explanation modality is evaluated by the fewest measures (of those included in the system)?

Evaluation Techniques:

- Competency questions, see Table 1 (modeling)
 - SPARQL queries
- Pellet reasoner (consistency)
 - Processed in 210ms
- OOPS! Ontology Pitfall Scanner (common design errors)
 - Issues from reused ontologies only
- *Future work*: schema for large knowledge graph (scaling/optimization)
- *Future work*: user testing (usability, general feedback)

Project Website:

